HACKOHI/O 2026

AI OUTAGE CONCIERGE

Predict. Explain. Prepare. Communicate.

A utility modernization challenge powered by anonymized outage damage, geospatial, network, and restoration data.

THE SCENARIO	THE USER	THE MISSION
A major storm is approaching	A customer seeking clear guidance	Turning complex operational data into trusted decisions

Challenge Overview

Design an AI-powered assistant that helps customers navigate storm-related outage uncertainty.

The Challenge

A major storm is approaching. Customers need timely, personalized, and understandable answers before an outage occurs and throughout restoration. Your team will build an AI-powered customer assistant that combines predictive analytics, geospatial context, network relationships, operational restoration inputs, and customer communication design.

The customer should be able to ask:

- "Will I lose power?"
- "How long could restoration take?"
- "What should I do to prepare?"
- "Do I qualify for outage alerts?"

Your Mission

- Estimate the likelihood of an outage for an anonymized premise, circuit, or location.
- Estimate a credible restoration time range using known or simulated damage and operational conditions.
- Provide clear preparation and safety recommendations tailored to the customer's context.
- Determine alert eligibility and recommend proactive communication and escalation triggers.
- Explain the key factors behind every prediction without exposing sensitive operational or customer data.

Why This Challenge Matters

Value	What success looks like
Customer impact	Customers receive useful guidance before, during, and after a storm.
Utility modernization	Teams connect AI, geospatial analysis, network topology, and restoration operations.
Operational realism	Recommendations reflect damage types, dependencies, labor, travel, and restoration sequencing.
Responsible AI	Predictions are transparent, privacy-conscious, safe, and appropriately uncertain.

Intended Users

Primary users are residential customers. Teams may also support anonymized priority locations such as life-support customers, water-treatment facilities, hospitals, schools, or other critical loads, provided the experience protects identity and avoids revealing sensitive locations.

Challenge Requirements

Build a working experience that demonstrates prediction, restoration reasoning, recommendation quality, and proactive communication.

1. Outage Risk Prediction

- Accept an anonymized premise, circuit, or map location as the customer context.
- Produce an outage-risk estimate or clearly defined risk band.
- Use relevant signals such as historical damage patterns, asset type, circuit or protective-device relationships, geospatial proximity, and available storm characteristics.
- Show the most important contributing factors and communicate uncertainty.

2. Restoration-Time Estimate

- Provide an estimated restoration duration or range rather than a false-precision promise.
- Consider damage type and quantity, upstream dependencies, step-restoration opportunities, labor-hour assumptions, travel, and nonproductive time.
- Update the estimate when damage, crew, access, or network assumptions change.
- Distinguish model output from confirmed utility restoration information.

3. Personalized Preparation Guidance

- Provide customer-friendly actions appropriate to the predicted risk and expected duration.
- Include safety, practical preparation guidance and clearly label emergencies or safety messages.
- Adapt recommendations for declared accessibility, medical, or critical-facility needs without inferring sensitive attributes.

4. Alerts and Proactive Communications

- Determine whether the scenario qualifies for outage-related alerts using documented rules or challenge assumptions.
- Recommend notification timing, channel, and escalation triggers.
- Generate concise, plain-language messages for pre-storm, outage-confirmed, estimate-updated, and restoration-complete scenarios.

5. Explainability and Responsible Use

- Make the reason for each recommendation understandable to a nontechnical customer.
- Disclose confidence, major assumptions, missing data, and conditions that could change the answer.
- Avoid exposing raw personal information, employee contact details, precise sensitive-facility locations, or internal-only operational fields.
- Include a clear fallback when the model cannot make a reliable prediction.

Challenge Data Package

The organizer-provided package should use sanitized, challenge-scoped extracts. Final availability may vary.

Core Data Sources

Data source	Potential fields or content	How teams may use it
ARCOS damage assessment	Anonymized outage or incident identifier, event, region, district, crew area, substation, circuit, report time, repair status and date, asset type, damage type, quantity, selected resource indicators, masked coordinates or geospatial key. (Shape file format)	Damage-pattern analysis, feature engineering, restoration-duration targets, spatial clustering.
Circuit GIS extracts	A set of selected circuits, anonymized circuit name identifiers, protective devices, and circuit relationships. (Shape file format)	Map experience, proximity signals, network context, visualization.
Upline flow / topology	Premise to upline equipment relationships and upstream dependencies. (TXT file)	Impact tracing, step restoration, dependency-aware prioritization.
STORMS labor-hour reference	Representative labor hours by damage type. (Excel spreadsheet)	Baseline restoration effort and scenario calculations.
Indirect labor assumptions	Representative travel and nonproductive-time multipliers, potentially averaged across operating companies. (Excel spreadsheet)	Operationally realistic duration estimates.
Priority-location indicators	Publicly available spatial and attribute data for hospitals, fire departments, and ambulance locations. (Shape file format)	Prioritization scenarios and tailored communications.
Structure diagrams	Sample distribution structure diagrams. (PDF)	Help to identify what materials might need to be replaced based on damage type.

Solution Expectations

Teams may choose their own architecture and modeling approach. The submission must make the end-of-the-end customer journey clear.

Minimum Viable Demonstration

1. A customer selects or enters an anonymized location.
2. The assistant returns an outage-risk prediction with a confidence indicator and explanation.
3. The customer explores damage or restoration scenarios and receives an estimated restoration range.
4. The assistant recommends preparation actions and indicates alert eligibility.
5. The solution generates at least one proactive customer communication.
6. The team shows how privacy, safety, uncertainty, and model limitations are handled.

Suggested Technical Components

Layer	Examples of acceptable approaches
Experience	Conversational interface, web or mobile prototype, map-based journey, accessible visualization.
Data	Curated tabular extract, geospatial layers, topology graph, and synthetic customer profiles were needed.
Analytics	Classification or risk scoring, regression or survival-style duration estimate, rules, graph analysis, geospatial features.
AI assistant	Retrieval and orchestration, natural-language explanation, scenario reasoning, communication generation.
Controls	Prompt and input validation, confidence thresholds, human escalation, privacy filters, audit trail.

Recommended Output Design

- Use a risk band and confidence level that are easy to understand.
- Present restoration as a range with the factors most likely to move it.
- Separate confirmed facts, model estimates, and customer actions.
- Use plain language and accessible interaction patterns.
- Allow judges to change at least one scenario input and observe the response.

Important

Teams are not expected to reproduce a production outage-management system. A focused, credible prototype with transparent assumptions is preferred over broad functionality that cannot be explained.